Exelon Generation.

Clinton Power Station
8401 Power Road
Clinton, IL 61727

U-604511
November 01, 2019

10 CFR 50.73
SRRS 5A.108

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Licensee Event Report 2019-001-01

Enclosed is Licensee Event Report (LER) 2019-001-01: Main Feed Breaker Failed to Close During Bus Shift. This report is being submitted in accordance with the requirements of 10 CFR 50.73.

There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. Dale Shelton Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

Thomas D. Chalmers
Site Vice President
Clinton Power Station

Attachment: Licensee Event Report 2019-001-01

cc:

Regional Administrator - Region III
NRC Senior Resident Inspector - Clinton Power Station
Office of Nuclear Facility Safety - Illinois Emergency Management Agency
NRC Project Manager, NRR - Clinton Power Station

IE22
NRR

NRC FORM 366
(04-2018)

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R 3 for instruction and guidance for completing this form http://www nrc gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
Clinton Power Station, Unit 1	05000461	1 OF 4

4. Title
Main Feed Breaker Failed to Close During Bus Shift

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number 05000
05	22	2019	2019	- 001 -	01	11	01	2019	Facility Name	Docket Number 05000

9. Operating Mode: 1

10. Power Level: 098

11. This Report Is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

☐ 20 2201(b)	☐ 20.2203(a)(3)(i)	☐ 50 73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20 2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50 73(a)(2)(ii)(B)	☐ 50 73(a)(2)(viii)(B)
☐ 20 2203(a)(1)	☐ 20 2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50 73(a)(2)(ix)(A)
☐ 20 2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50 73(a)(2)(iv)(A)	☐ 50 73(a)(2)(x)
☐ 20 2203(a)(2)(ii)	☐ 50 36(c)(1)(ii)(A)	☐ 50 73(a)(2)(v)(A)	☐ 73 71(a)(4)
☐ 20 2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50 73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20 2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73 77(a)(1)
☐ 20 2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50 73(a)(2)(v)(D)	☐ 73 77(a)(2)(i)
☐ 20 2203(a)(2)(vi)	☒ 50 73(a)(2)(i)(B)	☐ 50 73(a)(2)(vii)	☐ 73 77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact	Telephone Number (Include Area Code)
Mr. Dale Shelton, Regulatory Assurance Manager	(217) 937-2800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
X	EB	BKR	W	Y					

14. Supplemental Report Expected	15. Expected Submission Date	Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No				

Abstract (Limit to 1400 spaces, i.e , approximately 14 single-spaced typewritten lines)

On May 22, 2019, at 0925 CDT, at approximately 98.4% power while attempting to transfer 4160V bus 1A1 to the Reserve Auxiliary Transformer (RAT) (main) from the Emergency Reserve Auxiliary Transformer (ERAT) (reserve), RAT 4160V bus 1A1 main feed breaker 1AP07EK failed to close and immediately indicated tripped. A walkdown of feed breaker 1AP07EK showed no abnormal indications or relay flags. Feed breaker 1AP07EK was replaced and tested satisfactory. Initial troubleshooting identified that the latch check switch breaker closure permissive was not made up. As this permissive makes up following breaker closure it has been determined that feed breaker 1AP07EK was inoperable since April 17, 2019. The cause of the breaker failure was determined to be establishment of a breaker preventive maintenance strategy which was beyond the capability of the Molykote BR-2 lubricant used in the breaker. Changes to DHP Breaker preventive maintenance frequencies will be implemented to ensure that lubrication periodicities support the effective life of the lubricant. In addition, DHP Breaker maintenance and refurbishment procedures will be revised to incorporate use of Molykote G-4700 grease in place of Molykote BR-2. This event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as, "any operation or condition which was prohibited by the plant's Technical Specifications."

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Clinton Power Station, Unit 1	05000461	2019	- 001	- 01

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

General Electric -- Boiling Water Reactor, 3473 Megawatts Thermal Rated Core Power
Energy Industry Identification System (EIIS) codes are identified in text as [XX].

EVENT IDENTIFICATION

Main Feed Breaker Failed to Close During Bus Shift

A. Plant Operating Conditions Before the Event

Unit: 1	Event Date: May 22, 2019	Event Time: 0925
Mode: 1	Mode Name: Power Operation	Reactor Power: 098

B. Description of Event

On May 22, 2019, at 0925 CDT, at approximately 98.4% power while attempting to transfer 4160V bus 1A1 [BU] to the Reserve Auxiliary Transformer (RAT) (main) [XFMR] from the Emergency Reserve Auxiliary Transformer (ERAT) (reserve) [XFMR] in accordance with procedure CPS 3501.01, "High Voltage Auxiliary Power System," Section 8.1.8, RAT 4160V bus 1A1 main feed breaker 1AP07EK [BKR] failed to close and immediately indicated tripped. Annunciator 5060-1B [ANN] for auto trip of breaker was received. The operator placed the synchronizing switch in OFF prior to releasing the hand switch to the AUTO position in accordance with CPS 3501.01, Section 8.1.8.5, to prevent auto trip of the load breaker and the resulting loss of the bus.

A walkdown of feed breaker 1AP07EK showed no abnormal indications or dropped relay flags. The 4160V bus 1B1 [BU] was being supplied by the RAT with the Static Var Compensator (SVC) in service and showed no abnormal indications. The 4160V bus 1C1 [BU] was being supplied by the ERAT with the SVC in service and showed no abnormal indications. Feed breaker 1AP07EK was last closed on April 17, 2019, during performance of procedure CPS 9080.01, "Diesel Generator 1A Operability - Manual and Quick Start Operability Surveillance." Initial troubleshooting identified that the latch check switch breaker closure permissive was not made up. As this permissive makes up following breaker closure it has been determined that feed breaker 1AP07EK had not been operable since its last closure on April 17, 2019.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Clinton Power Station, Unit 1	05000461	2019	- 001	- 01

NARRATIVE

C. Cause of the Event

The Preventive Maintenance strategy for DHP breakers at CPS consists of an eight-year breaker preparation and swap activity and a twenty-year vendor refurbishment activity. These frequencies take into consideration performance history, manufacturer recommendations, and Performance Centered Maintenance (PCM) guidance. The eight-year preparation and swap activity was based on the mild operating environment, three to ten year guidance in the PCM Template Bases, and a manufacturer recommendation of once per 2000 cycles. The failure analysis conducted by Westinghouse for the breaker identified that lubrication throughout the breaker was found to be dry or tacky. Therefore, the cause of the feed breaker 1AP07EK failure to close was determined to be establishment of a breaker preventive maintenance strategy which was beyond the capability of the Molykote BR-2 lubricant used in the breaker.

D. Safety Consequences

There were no safety consequences associated with the event described in this report as the affected bus remained energized from the reserve feed (ERAT). If the ERAT had become inoperable the DGs were available to start and provide power to the bus. Because Limiting Condition for Operation (LCO) 3.8.1, "ELECTRICAL POWER SYSTEMS – AC Sources – Operating," Condition A, was not entered from April 7, 2019 until the condition was discovered on May 22, 2019, this event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as, "any operation or condition which was prohibited by the plant's Technical Specifications."

E. Corrective Actions

(1) Feed breaker 1AP07EK was replaced and tested satisfactory.

(2) Changes to DHP Breaker preventive maintenance frequencies will be implemented to ensure that lubrication periodicities support the effective life of the lubricant.

(3) DHP Breaker maintenance and refurbishment procedures will be revised to incorporate use of Molykote G-4700 grease in place of Molykote BR-2.

F. Previous Similar Occurrences

There were no previous events identified involving a breaker failure similar to the occurrence described in this licensee event report.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Clinton Power Station, Unit 1	05000461	2019	- 001	- 01

NARRATIVE

G. Component Failure Data

Manufacturer: Westinghouse
Component Type: DHP 350, 4160 Volt Power Circuit Breaker